Time: 1hr. Set-2 CT-1, 2025 BE-IT 2nd Year 1st Semester Subject: Mathematics for IT-II (IT/BS/B/T/222) Full Ma

CO1	a. For any two events A and B, prove that $P(AB) \leq P(A+B) \leq P(A)+P(B)$. b. In a test, an examinee either guesses or copies or knows the answer to a multiple choice question with four choices. The probability that he makes a guess is 1/3 and the probability that he copies the answer 1/6. The probability that his answers is correct given that he copied is 1/8. Find the probability that he knew the answer to the question given that he correctly answered it. c. Is the following a pdf? $f(x) = 2x,\ 0<x\leq 1;\ 4-2x,\ 1<x\leq 2;\ 0$, elsewhere.	 3
CO2	a. Derive the expressions of mean and variance of Poisson distribution with given parameter. $z = \frac{x-\mu}{\sigma}$ b. A distribution has mean 10.4 and s.d. 1.2. Find the limits within which an observation chosen at random almost certainly lies. c. 5000 candidates appeared in an examination, in which the minimum for a pass is 40 and the minimum for a distinction is 50. If it is a known that the average mark obtained by the candidates is 43 and the S.D. is 7. Find how many of the candidates expect to get simply "pass" and the number obtained distinction. Given $\varphi(0.43) = 0.6664$, $\varphi(1) = 0.8413$.	3+ 3 3
CO3	$\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$ a. Show that the mean and standard error of sample mean $(\bar{x})$ from the samples of size n are $E(\bar{x}) = \mu$ and $S.E.(\bar{x}) = \sigma/\mu^{1/2}$ $\bar{x}-\mu$ b. A normal population has a mean 0.1 and a standard deviation 2.1. Find the probability that the mean of a sample of size 900 will be negative. It is given that the probability that the absolute value of a standard normal variate exceeds 1.43 is 0.153	4

Time: 1hr. Set-3 CT-1, 2025 BE-IT 2nd Year 1st Semester Subject: ...

CO1	a. If events A and B are independent then prove that A' and B' are also independent.	4
	b. A bag contains 1 B and 2 W balls. A drawing from a bag consists of taking A ball from the bag and keeping it out of it is W but putting it back if it is B. Calculate the probabilities i) the first drawing is a W ball, ii) the second drawing is a W ball and iii) the third drawing is a W ball.	3
	c. The length of life X (in hours) of a certain electronic components is assumed to follow a continuous distribution with density function $f(x)=k/x^2$, $(1000 \leq x \leq 1500)$. Determine the constant k and find the probability that a component selected at random will function for at least 1200 hours.	
CO2	a. Approximate i) Poisson distribution from Binomial distribution and ii) Normal distribution from Poisson distribution.	3+3
	b. If 3% of the bolts manufactured by a company are defective, what is the probability that in a sample of 200 bolts, 5 will be defective? (Given $e^{-6} = 0.00248$).	3
	c. A die is tossed 1200 times. Find the probability that the number of "sixes" lies between 190 and 210. (Given that area under the standard normal curve between z=0 and z=0.78 is 0.2823 and z=0 to z=0.81 is 0.2910.)	3
CO3	a. Prove that $\sum_{i=1}^{n}(x_i-A)^2$ is the least when $A = \bar{x}$, where $x_1, x_2, \ldots x_n$ are the observations, A is any arbitrary constant and $\bar{x}$ is the arithmetic mean.	4
	b. The variable x is normally distributed with mean 68 inches and s.d. 2.5 inches. What is the size of the sample whose mean shall not differ from the population mean by more than one inch, with probability 0.95? (Given that area of under standard normal curve to the right of the coordinate at 1.96 is 0.025)	4

Time: 1hr. <u>Set-1</u> **CT-1, 2025 BE-IT 2^{nd} Year 1^{st} Semester Subject: Mathematics for IT-II (IT/BS/B/T/222) Full Marks: 30**

CO1	a. If events A and B are not mutually exclusive then show that $P(AB) \geq P(A)+P(B) -1$.	3
	b. In a bolt factory machine M_1, M_2 and M_3 manufacture respectively 25%, 35% and 40% of the total of their output with 5%, 4%, and 2% respectively are defective bolts. One bolt is chosen at random from the product and it is found defective. What are the probabilities that it was manufactured in machine M_1 and M_3?	4
	c. Given the pdf $f(x) = (\frac{1}{2})\, e^{-x/2}, x>0$; evaluate the distribution function.	3
CO2	a. Derive the expressions of mean and variance of Binomial distribution with given parameters	3+3
	b. Player A tosses 5 coins and player B tosses 8 coins. If the coins are unbiased, what is the probability of obtaining a total of 6 heads by the two players?	3
	c. Assume the mean height of soldiers to be 68.22 inches with a variance of 10.8 sq. inches. How many soldiers in a regiment of 1000 would you expect to be over 6 ft. tall? (Given that area under the standard normal curve between x=0 and x=0.35 is 0.1368 and between x=0 to x=1.35 is 0.3749.)	3
CO3	a. n pairs of values of two variables x and y are given. The variances of x, y and (x-y) are σ_x^2, σ_y^2 and σ_{x-y}^2 respectively. Show that the correlation coefficient r_{xy} between x and y is $(\sigma_x^2+\sigma_y^2-\sigma_{x-y}^2)/2\sigma_x\sigma_y$.	4
	b. A random sample of size 25 is taken from a $N(\mu, \sigma^2)$ with $\mu=30$ and $\sigma^2=16$. Would the probability that the sample mean would lie between 25 and 35 be greater than 0.99?	4

$2^{14} \geq 1024 \times 8$